

Evidence of Phase Transitions in Small Transformer-Based Language Models

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ABSTRACT Phase transitions have been proposed as the origin of emergent abilities in large language models (LLMs), where new capabilities appear abruptly once models surpass critical thresholds of scale. Prior work, such as that of Wei et al., demonstrated these phenomena under model and data scaling, with transitions revealed after applying a log scale to training compute. In this work, we ask three complementary questions: (1) Are phase transitions unique to large models, or can they also be observed in small transformer-based language models? (2) Can such transitions be detected directly in linear training space, rather than only after log rescaling? and (3) Can these transitions emerge at early stages of training? To investigate, we train a small GPT-style transformer on a character-level corpus and analyze the evolution of vocabulary usage throughout training. We track the average word length, the number of correct versus incorrect words, and shifts in vocabulary diversity. Building on these measures, we apply Poisson and sub-Poisson statistics to quantify how words connect and reorganize. This combined analysis reveals a distinct transition point during training. Notably, these transitions are not apparent in standard loss or validation curves, but become visible through our vocabulary- and statistics-based probes. Our findings suggest that phase-transition reorganizations are a general feature of language model training, observable even in modest models, detectable directly in linear training space, and occurring surprisingly early as coherence emerges. This perspective provides new insight into the nonlinear dynamics of language model training and underscores the importance of tailored metrics for uncovering phase transition behaviors.

INDEX TERMS Transformer, Phase transition, Emergent abilities, Poisson statistics, KL divergence, Language models

I. INTRODUCTION

Frontier Large language models (LLMs) have demonstrated striking emergent capabilities in reasoning, coding, and multimodal understanding [1]–[5]. These breakthroughs have been widely associated with scaling: as models grow in parameter count, dataset size, and training compute, they appear to acquire qualitatively new abilities. Wei et al. characterized such emergent behaviors as phase transitions, highlighting that they often arise abruptly at critical thresholds rather than increasing smoothly with scale [6].

Despite this progress, most prior studies analyze phase transitions indirectly, requiring billion-parameter models and reporting results only after applying logarithmic rescaling of compute or training steps. As a result, it remains unclear whether phase-transition-like reorganizations can be observed in small, tractable models rather than only in massive LLMs, and raises the question whether such

discontinuities can be detected directly on a linear training axis without resorting to log transformations.

Addressing this gap is important for two reasons. First, small models serve as practical laboratories: they can be trained and inspected with modest resources while still revealing fundamental learning dynamics. Second, if phase transitions are observable directly in raw training steps, this strengthens the case that emergent behaviors are an intrinsic property of neural training, not merely an artifact of evaluation metrics or log scaling. Such insights can inform both interpretability research and the design of more efficient models.

In this work, we investigate phase-transition dynamics in a small GPT-style transformer (3.6M parameters) trained with character-level tokenization on the Tiny Shakespeare corpus. To detect discontinuities, we introduce a Poisson-centered diagnostic framework [7] that tracks dispersion,

Kullback–Leibler (KL) divergence, vocabulary composition, and word length. These statistical probes reveal synchronized discontinuities in training that coincide with a transition from random, fragmentary sequences to coherent words. We argue that lexical-level coherence functions as an order parameter for emergent linguistic ability.

The remainder of this paper is organized as follows. Section II reviews related work on emergent abilities, phase transitions, discontinuous training dynamics, and statistical probes. Section III describes the experimental setup and diagnostic methodology. Section IV presents our findings on synchronized discontinuities. Section V discusses implications, limitations, and directions for future work.

II. RELATED WORK

Our investigation of phase transitions in small transformer-based language models draws on three interconnected research traditions: the physics of phase transitions and collective phenomena, the empirical discovery of emergent abilities in large language models, and mechanistic studies of discontinuous learning dynamics. We review each tradition in turn, highlighting how they converge to motivate our central questions about the observability, detectability, and timing of phase-transition-like reorganizations in modest-scale systems.

A. PHASE TRANSITIONS IN STATISTICAL PHYSICS

The formal study of phase transitions originates in statistical mechanics, where smooth microscopic laws give rise to singular macroscopic behavior. Landau and Lifshitz’s canonical treatment establishes the foundational concepts: an order parameter—a macroscopic quantity characterizing the state of the system—changes abruptly as a control parameter crosses a critical threshold [8]. In second-order (continuous) transitions, the order parameter evolves smoothly from zero to finite value, accompanied by diverging fluctuations and correlation lengths. In first-order (discontinuous) transitions, the order parameter jumps discontinuously, reflecting the coexistence and exchange of distinct phases. Crucially, these qualitative reorganizations emerge from collective interactions among many components, not from singularities in the underlying microscopic rules. The water–vapor transition provides an intuitive example: as temperature or pressure varies, the system’s density undergoes a sharp, measurable discontinuity despite the smoothness of intermolecular forces.

Tong’s modern pedagogical exposition extends this framework through renormalization group (RG) theory and effective field descriptions [9]. RG analysis reveals how critical behavior is governed by fixed points in the space of couplings, organizing transitions into universality classes defined by shared scaling exponents rather than microscopic details. Tong emphasizes that apparent macroscopic discontinuities arise from the amplification of microscopic fluctuations through many-body correlations—a perspective that has proven fruitful when transferred to high-dimensional

learning systems. His treatment of effective potentials, in which the system’s state concentrates near minima of a coarse-grained free energy, provides a conceptual template later adopted by theorists modeling neural network training as navigation through an effective landscape of competing representations.

Together, Landau–Lifshitz and Tong establish a rigorous vocabulary for describing qualitative reorganizations: order parameters, control parameters, critical thresholds, coexistence regimes, and scaling laws. Although developed for physical systems in thermal equilibrium, this language has migrated to describe abrupt changes in artificial systems, including neural networks, where “temperature” may be replaced by training time, learning rate, or model scale, and “magnetization” by generalization accuracy, representational coherence, or statistical dispersion.

B. EMERGENCE AS A GENERAL PRINCIPLE IN COMPLEX SYSTEMS

The conceptual leap from physics to broader contexts was articulated influentially by Anderson, who argued that “more is different”—that increasing complexity introduces qualitatively new organizational principles not reducible to component-level laws [10]. Using examples from superconductivity to molecular biology, Anderson asserted that each level of scale in a hierarchy exhibits emergent properties and demands its own effective theories. This essay laid the philosophical groundwork for studying emergent computation, collective intelligence, and self-organization across disciplines. For artificial intelligence, Anderson’s thesis legitimizes the expectation that large-scale neural systems may undergo abrupt reorganizations in capability or structure that cannot be predicted from isolated neuron dynamics or gradient updates.

Building on this foundation, Huberman and Hogg demonstrated that artificial intelligence systems—specifically, symbolic search and constraint-satisfaction algorithms—can exhibit sharp threshold phenomena as control parameters vary [11]. They analyzed how problem difficulty peaks near phase boundaries between solvable and unsolvable regimes, analogous to critical slowing down in physics. Their work established that combinatorial and computational systems, even far from thermodynamic equilibrium, display transition-like behaviors where collective properties change abruptly. Although their focus was symbolic AI rather than deep learning, the conceptual parallel is clear: large computational systems can have critical regions where behavior reorganizes discontinuously.

Forrest extended these ideas into the realm of emergent computation in distributed systems, including early neural networks and evolutionary algorithms [12]. She argued that computation itself can be an emergent property arising from local interactions that self-organize into global structure. By reviewing cellular automata, associative memories, and genetic algorithms, Forrest highlighted how non-linear coupling of simple rules produces persistent patterns and adap-

tation. While largely qualitative and pre-dating modern deep learning, her synthesis situates neural training within the broader study of complex adaptive systems, reinforcing the expectation that emergent coherence in language models—such as the abrupt acquisition of multi-character words we observe—can arise from smooth, distributed updates.

C. PHASE TRANSITIONS IN NEURAL NETWORKS: EARLY THEORETICAL FOUNDATIONS

The application of phase-transition concepts directly to neural networks began in earnest in the 1990s. Kinzel's review article provided one of the first systematic treatments, analyzing how simple learning systems—perceptrons, Hopfield networks, and spin-glass models—exhibit threshold behavior [13]. For instance, a perceptron trained on random patterns transitions sharply from an unlearnable to a learnable regime once the ratio of training examples to weights exceeds a critical value. Similarly, associative memory networks undergo order-disorder transitions when capacity limits are exceeded, with retrieval accuracy collapsing abruptly beyond a threshold. Kinzel connected these findings to energy-landscape reorganization and symmetry breaking familiar from condensed-matter physics. His work demonstrated that learning dynamics naturally partition into distinct regimes—ordered (generalizing) and disordered (memorizing or random)—separated by sharp boundaries. While the models studied were low-dimensional and analytically tractable, Kinzel's analysis established that phase-transition language applies meaningfully to neural computation, foreshadowing later efforts to describe deep learning through statistical mechanics.

In the modern era, Bahri and colleagues have synthesized a decade of research applying statistical-mechanical tools to deep neural networks [14]. Their comprehensive review organizes theory around three axes: the geometry of high-dimensional loss landscapes, the dynamics of signal propagation and criticality (including “edge-of-chaos” regimes where networks balance order and chaos), and scaling laws linking architecture, dataset size, and generalization. They describe how random-matrix theory and mean-field approximations reveal phase-transition-like separations between ordered and chaotic signal-flow regimes, and how such transitions influence trainability. Importantly, Bahri et al. caution that deep learning operates far from thermodynamic equilibrium; hence, “phase transition” serves as a metaphor for qualitative regime changes in optimization dynamics rather than literal thermodynamic phenomena. Nonetheless, the conceptual apparatus—order parameters, critical points, universality—provides a powerful framework for interpreting abrupt shifts in network behavior. For our study, Bahri et al. supply rigorous justification for treating discontinuities in dispersion or KL divergence as emergent reorganizations of representation statistics.

Miller, Zhang, and Ganguli have recently proposed that deep networks trained by gradient descent can be modeled as non-equilibrium systems undergoing absorbing-state

phase transitions [15]. They define an order parameter based on the fraction of active degrees of freedom in weight space and demonstrate that training dynamics approach an absorbing state where updates cease, analogous to the frozen phase in directed-percolation universality classes. Empirical scaling exponents measured on feed-forward and recurrent architectures match those predicted by non-equilibrium statistical physics, suggesting universality across network types. While highly theoretical and not linguistically oriented, their work provides formal grounding for describing sharp onsets of regularization—such as the sub-Poisson stabilization we observe—as regime transitions in high-dimensional optimization.

D. EMERGENT ABILITIES IN LARGE LANGUAGE MODELS

The empirical discovery of discontinuous capability gains in large-scale language models has catalyzed recent interest in phase-transition analogies. The GPT series documents this progression. Radford et al. introduced GPT-1, demonstrating that unsupervised generative pretraining on a large text corpus followed by supervised fine-tuning yields transferable representations that substantially outperform models trained from scratch [1]. The key insight was that a single, generic language model can adapt to many downstream tasks with minimal labeled data, suggesting that broad linguistic structure emerges from next-token prediction alone. GPT-2 scaled this paradigm to 1.5 billion parameters and showed that zero-shot multitask performance arises spontaneously without explicit fine-tuning [2]. Capabilities such as translation, summarization, and question answering appeared organically, motivating the notion that sufficient scale and data diversity induce qualitative leaps in coherence and reasoning.

Brown et al.'s GPT-3, at 175 billion parameters, established scaling as the dominant driver of emergent generalization [3]. While training loss followed smooth power laws, task performance exhibited abrupt gains in reasoning, arithmetic, and translation once model size exceeded tens of billions of parameters. The paper introduced few-shot, one-shot, and zero-shot evaluation protocols, showing that in-context learning—without gradient updates—becomes viable at scale. GPT-3 thus provided large-scale empirical evidence that internal representations cross thresholds beyond which qualitatively new behaviors emerge, paralleling the discontinuities we observe in miniature systems.

Subsequent frontier models reinforced this pattern. Chowdhery et al.'s PaLM, at 540 billion parameters, demonstrated emergent reasoning at unprecedented scale, with performance curves showing discontinuous jumps in logical and multi-step tasks once critical thresholds were surpassed [4]. The authors attributed improvements to both raw scale and dataset diversity, noting that emergent capabilities correlate with cross-domain exposure. Similarly, OpenAI's GPT-4 Technical Report documented marked improvements in reasoning, coding, and multimodal comprehension, with